S1 Text.

ODD description of EvoSim

Supporting information for "Environmental fluctuation strengthens gene regulation only when it can be sensed"

This description follows the ODD protocol in its 2020 revision (Grimm et al., J Artif Soc Soc Simul 23(2):7). It documents the version of EvoSim used for the experiments in the main text, and every parameter value is given in Table 1. Where a mechanism is switched off in these experiments it is described briefly and marked as unused.

1. Purpose and patterns

EvoSim is an individual-based model of unicellular organisms whose genomes are gene regulatory networks of evolvable size and connectivity. In this study it is used to ask how environmental fluctuation, and the information available to track it, affect the number of regulatory connections and their strength. The model is built so that network size and connectivity evolve under mutation, selection and drift. Each phenotypic trait has a cost as well as a benefit, so traits settle at interior values, and the resource ecology is built partly by the organisms themselves through the byproducts of consumption.

2. Entities, state variables and scales

Grid cells. The environment is a square lattice of $G \times G$ cells ($G = 128$ in this study) with reflecting boundaries. Each cell holds the concentration of six resource channels, $R_r(x, y)$ for $r = 1, \ldots, 6$, capped at 6.0, and a count of the organisms occupying it (at most 8).

Organisms. Each organism has a position, an energy store (at most 4.0), an age in time steps, a lineage depth (number of ancestral divisions), a deleterious-load counter, and a genome. The genome is a regulatory network with n_g nodes, $13 \leq n_g \leq 32$, consisting of a weight matrix $W \in \mathbb{R}^{n_g \times n_g}$ and a bias vector $b \in \mathbb{R}^{n_g}$. Nodes 1–12 are the phenotype outputs, node 13 is the sensor for the abiotic condition, and nodes $14, \ldots, n_g$ are hidden nodes with no fixed meaning. Each organism also carries its phenotype, computed once at birth and fixed for life in these experiments. It consists of twelve traits in $[0, 1]$, namely six resource-uptake shares (normalised to sum to one), body size, motility, division threshold, DNA-repair fidelity, stress tolerance and leakage.

Global state. The abiotic condition, which cycles through discrete states (Section 7.3); the seasonal phase; and a famine counter.

Scales. Time advances in discrete steps. In the experiments of this study one generation took about 24 time steps in fluctuating and 31 in frozen populations (measured from the slope of mean lineage depth over the second half of each run), so a 150,000-step run covers roughly 5,000 to 6,000 generations. The maximum age is 60 steps.

3. Process overview and scheduling

Each time step executes the following steps in order.

1. *Resource renewal.* Every channel in every cell decays, and external inflow is added (Section 7.1).

2. *Shock check.* With a small probability a mass-mortality or famine event occurs (Section 7.9).

3. *Reversible development* (Section 7.4), which was not used in this study.

4. *Organism update.* Organisms are visited once each, in storage order. For each organism the step comprises uptake and conversion of resources, with byproduct release into the cell (Section 7.5); maintenance cost (7.6); energy update; ageing unless dormant (7.7); movement (7.8); death if energy is exhausted or the maximum age is exceeded; otherwise, if the energy store exceeds the division threshold and the organism is not dormant, it is queued to divide.

5. *Births* (Section 7.10). Queued parents are processed in queue order. Placement is serial; mutation and development of the offspring are computed in parallel with a random seed drawn serially for each offspring, so results do not depend on the number of threads.

6. *Diffusion* of all channels (Section 7.2).

Because organisms are updated sequentially and draw on shared cell resources, an organism visited earlier in a step can deplete a cell before a later one reaches it. Storage order reflects the reuse of memory slots and has no biological meaning.

4. Design concepts

Basic principles. Every trait that brings a benefit also carries a cost, so that selection does not drive traits to a boundary. Uptake is an allocation of a fixed budget across the six channels, so the affinities are not independent. Consumption of channel r releases a byproduct into channel $r + 1$, so all channels except the first exist only through the activity of the organisms.

Emergence. Network size, connectivity and connection strength, population size, resource distribution and the composition of the community emerge from the individual processes.

Adaptation. Organisms do not adapt within their lifetime. The phenotype is fixed at birth and is a function of the genome, the local resource concentrations and the value of the abiotic cue at the place and time of birth.

Objectives, learning and prediction. None; there is no explicit fitness function. Reproductive success follows from energy balance.

Sensing. At birth, an organism senses the concentrations of the six channels in its cell and the value of the abiotic cue.

Interaction. Organisms interact through shared resources, through byproducts that others can consume, through crowding costs among organisms in the same cell, and through the occupancy limit that can block a division.

Stochasticity. Mutation, movement, the placement of offspring, shocks and the founding genomes are random. All randomness derives from the simulation seed, and runs are exactly reproducible for a given seed.

Collectives. None are modelled explicitly. Lineages are tracked as clades for bookkeeping only; they do not affect the dynamics.

Observation. Population means of the phenotype, of network structure (number of nodes, number of non-zero weights, mean absolute weight) and of ecological quantities are recorded at every time step. The measures used in the main text are defined in Section 7.12.

5. Initialisation

All cells start with $R_1 = 1.0$ and the other channels empty. $\max(200, G^2/80)$ founders (204 at $G = 128$) are placed in uniformly random cells with energy 0.8, age 0 and load 0. Each founder has $n_g = 16$ nodes; each weight is non-zero with probability 0.25 and then drawn from $\mathcal{N}(0, 0.6^2)$, and biases are drawn from $\mathcal{N}(0, 0.2^2)$. Founders are developed in the conditions of their cell before the first step.

6. Input data

None. The model uses no external data.

7. Submodels

7.1 Resource renewal

Each step, every channel in every cell is multiplied by $1 - 0.006$. External inflow per cell is

$$I(t) = 0.25 \left(1 + 0.6 \sin \frac{2\pi t}{15{,}000}\right) \times f(t),$$

where $f(t) = 0.65$ during a famine and 1 otherwise, and $I(t)$ is set to zero if negative. In frozen environments (one condition state) all inflow enters channel 1. In fluctuating environments 60% enters channel 1 and 40% enters the channel with the highest yield in the current state (Section 7.3). Concentrations are capped at 6.0 after each addition.

7.2 Diffusion

Each channel diffuses by a five-point Laplacian with rate 0.12 per step and reflecting (no-flux) boundaries; concentrations are capped at 6.0.

7.3 Abiotic condition, state and cue

The condition has n_s states ($n_s = 6$ in fluctuating arms, 1 in frozen arms) and a period of $P = 2{,}000$ steps. With phase $\theta(t) = (t \bmod P)/P$, the current state is $s = 1 + \lfloor n_s \theta \rfloor$. The state rotates the yields, so that the yield of channel r in state s is $y_{1+((r-1+s-1) \bmod 6)}$, with $y = (1.00, 0.60, 0.45, 0.35, 0.28, 0.22)$, so a different channel pays best in each state. The cue delivered to node 13 is defined as follows.

- In the informative arm $c = 2\theta - 1$, which determines the state exactly.
- In the yoked arm $c = 2\phi - 1$, with ϕ the fractional part of $(t \bmod P) \cdot 6.1803399/P$, an incommensurate period (the multiplier is $5 + \varphi$, with φ the golden ratio), which covers the same range but carries 0.0004 bits of information about the state.
- In the no-cue arm $c = 0$.

With $n_s = 1$ the state is always 1 and the cue is zero, so the three frozen arms are identical.

7.4 Development

At birth the expression vector is initialised to $e_k = 0.1$ and updated synchronously eight times,

$$e_k \leftarrow \tanh\Big(b_k + \sum_{m=1}^{n_g} W_{mk}\, e_m + u_k\Big),$$

where $u_k = R_k$ at the organism's cell for $k \le 6$, $u_{13} = c$, and $u_k = 0$ otherwise. The phenotype is $\phi_k = (e_k + 1)/2$ for $k = 1, \ldots, 12$, and the uptake shares $\phi_1, \ldots, \phi_6$ are rescaled to sum to one. Development does not reach a fixed point in general; the phenotype is the state after the eighth update, which is deterministic and heritable. Reversible development, in which living organisms are re-developed every few steps, exists in the model but was not used in this study.

7.5 Uptake and conversion

Let $\ell = 0.45 + 0.55\,\phi_{12}$ be the effective leakage and $m = (0.5 + \phi_7)(1 + 1.2\,\ell)$ the rate multiplier (multiplied by 0.35 when dormant). For each channel r with share $a_r = \phi_r > 0$ and local concentration R_r, the amount taken is $g_r = \min(0.6\,R_r\,a_r\,m,\ R_r)$. It is removed from the cell, the organism gains $g_r\,y'_r\,(1 - \ell)$ energy, where y'_r is the current yield of channel r, and $g_r\,\ell$ is added to channel $r + 1$ in the same cell (the byproduct of channel 6 leaves the system). The summed concentration of channels 2–6 in the cell is recorded as the local toxicity T.

7.6 Maintenance

The energy cost per step is

$$0.08 + 0.22\,\phi_7^2 + 0.035\,\phi_8 + 0.04\,\phi_{11}^2 + 0.02\,\phi_{10} + 0.0015\,(n_g - 13) + \kappa_e\,n_{nz} \\ + 0.05\,(\text{other organisms in the cell}) + 0.015\,L + 0.008\,T\,(1 - \phi_{11}),$$

where n_{nz} is the number of non-zero weights, $\kappa_e = 5 \times 10^{-5}$ in these experiments, and L is the load counter. The cost is multiplied by 0.05 when dormant. Energy is updated as gains minus cost, capped at 4.0.

7.7 Dormancy and ageing

An organism whose energy at the start of the step is below 0.15 is dormant for that step. A dormant organism has reduced uptake (Section 7.5) and a reduced maintenance cost (Section 7.6), does not age and cannot divide. Otherwise its age increases by one.

7.8 Movement

With probability ϕ_8 the organism moves to a cell chosen uniformly from its Moore neighbourhood including its own cell, clamped at the boundary.

7.9 Shocks

At each step, with probability 1/25,000, one of three events occurs, each with probability 1/3: a cull, in which every organism dies with probability 0.92; a heat shock, in which each organism dies with probability $0.92\,(1 - \phi_{11})$; or a famine, which multiplies inflow by 0.65 for the next 400 steps. These events occur in all arms.

7.10 Division and inheritance

An organism that is not dormant and whose energy is at least $2.0\,(0.5 + \phi_9)$ divides. The offspring is placed in a random cell of the parent's Moore neighbourhood (including the parent's cell); if that cell already holds 8 organisms the division fails. Parent and offspring each receive half the parent's energy. The offspring has age 0, lineage depth one greater than the parent, and load equal to the parent's load plus the change from mutation (Section 7.11), floored at zero. Its genome is a mutated copy of the parent's, and it is developed in the conditions of its own cell.

7.11 Mutation

The per-site rate is $\mu = \max\left(10^{-3}(1 - 0.85\,\phi_{10}^{\text{parent}}),\ 10^{-6}\right)$, so DNA repair can reduce it by up to 85%. At each birth the offspring genome receives a Poisson number of point mutations with mean μn_g^2, each adding $\mathcal{N}(0, 0.25^2)$ to a randomly chosen weight, and with probability μn_g one randomly chosen bias receives $\mathcal{N}(0, 0.25^2)$ as well. With probability 0.2 (in these experiments; 0 by default) one randomly chosen weight is set to exactly zero. We call this connection silencing. Without it, additive mutation would drive every weight away from zero and density towards one.

Structural mutations act on whole nodes. With probability 0.002 all outgoing weights of a random node are set to zero (gene silencing). With probability 0.002 the incoming and outgoing weights and the bias of a random node are copied onto another random node, and $\mathcal{N}(0, 0.05^2)$ noise is added to every copied weight, including weights that were zero in the source (gene duplication). With probability 0.003 a hidden node is added, if $n_g < 32$, as a copy of a random existing node made in the same way as a duplication. If no node is added, a random hidden node is removed with probability 0.003, provided $n_g > 13$. In addition, each point mutation, bias mutation and connection silencing is classified for the load counter as deleterious (probability 0.70, load $+1$), beneficial (0.05, load -1) or neutral (otherwise), and each gene silencing adds 1 to the load. Load carries a maintenance cost (Section 7.6) and can fall as well as rise.

7.12 Measures used in the main text

Two quantities are computed for each organism at birth, the number n_{nz} of weights with $|w| > 10^{-4}$ and the strength density $\sum |w|/n_g^2$ over all n_g^2 weights, zeros included. At each time step the population averages of count density n_{nz}/n_g^2, strength density and total strength $\sum |w|$ (strength density times n_g^2) are recorded. Each run is summarised by the mean over its final 20% of time steps, with ratios computed per step before averaging.

Parameters

Table 1. EvoSim parameters. "Default" is the value in the code; "This study" lists values that differ from the default.

Parameter	Default	This study
Grid size G	256	128
Maximum organisms per cell	8	
Founders	$\max(200, G^2/80)$	204
Base inflow	0.25	
Season period, amplitude	15,000; 0.6	
Decay per step	0.006	
Diffusion rate	0.12	
Resource cap	6.0	

Parameter	Default	This study
Yields $y_1, \ldots, y_6$	1.00, 0.60, 0.45, 0.35, 0.28, 0.22	
Condition states n_s	1	6 (fluctuating), 1 (frozen)
Condition period P	2,000	
Inflow share following the favoured channel	0	0.4 (fluctuating)
Cue gain	1	0 (no-cue arms)
Yoked cue	off	on (yoked arms)
Uptake rate	0.60	
Leakage floor; rate bonus	0.45; 1.2	
Maintenance per step (base, size^2, motility, stress^2, repair)	0.08, 0.22, 0.035, 0.04, 0.02	
Maintenance per hidden node	0.0015	
Maintenance per non-zero weight κ_e	0	5×10^{-5}
Maintenance per step (crowding, load, toxicity)	0.05, 0.015, 0.008	
Division energy	2.0	
Initial energy; maximum energy	0.8; 4.0	
Dormancy threshold; upkeep; uptake	0.15; 0.05; 0.35	
Maximum age	60	
Development iterations	8	
Per-site mutation rate; σ	10^{-3}; 0.25	
Maximum repair suppression	0.85	
Gene silencing; duplication	0.002; 0.002	
Node addition; deletion	0.003; 0.003	
Connection silencing per birth	0	0.2
Deleterious; beneficial mutation share	0.70; 0.05	
Initial nodes	16	
Shock probability per step; kill fraction	1/25,000; 0.92	
Famine length; inflow factor	400; 0.65	
Run length		150,000 steps
Seeds		1–30 per arm